are headed toward gargantua, swings the rays around itself, and ejects them toward the camera. This explains the donut of starlight that surrounds the IMBH's shadow. Although the IMBH is a thousand times smaller than Gargantua, it is far closer to the Ranger than is Gargantua, so it looks only modestly smaller.

As the IMBH appears to move rightward, as seen by the slingshot-moving camera, it leaves Gargantua's primary shadow behind itself (middle picture in Figure 8.7), and it pushes a secondary image of Gargantua's shadow ahead of itself. These two images are completely analogous to the primary and secondary images of a star gravitationally lensed by a black hole; but now it is Gargantua's shadow that is being lensed, by the IMBH. In the bottom picture, the secondary shadow is shrinking in size, as the IMBH moves onward. By this time the slingshot is nearly complete, and the camera, on board the Ranger, is headed downward, toward Miller's planet.

As impressive as these images may be, they can be seen only up close to the IMBH and Gargantua, not from the great distance of Earth. To astronomers on Earth, the most visually impressive things about gigantic black holes are jets that stick out of them and the light from brilliant disks of hot gas that orbit them. To these we'll now turn.

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20 See Figures 6.4 and 6.5.